

Aerolytics: What Makes Our Project Unique?

1. The Problem with Existing Apps

Most air quality apps today only do two simple things: they tell you what the pollution level is right now, and maybe they guess what it will be tomorrow.

But they miss the most important question: **Why is the pollution bad today?**

Older apps try to guess the "why" using something called *correlation*. Correlation just means two things happen at the same time. For example, if it gets cold and pollution goes up, an older app might say the cold *caused* the pollution. But that's often wrong—they just happened together by coincidence.

The main thing that makes **Aerolytics** unique and interesting is that we completely fixed this problem using a new module called the **Causality Engine**.

2. The Big Novelty: The Causality Engine

Our project stands out because we built a math engine that proves exactly what is driving the pollution. We moved away from simple guessing and used a rigorous statistical method called **Granger Causality**.

How does it work?

Instead of just looking at what is happening right now, our system looks back in time. It asks a strict mathematical question: *If we look at the traffic emissions (NO2) from 3 hours ago, does it perfectly predict the PM2.5 dust in the air right now?*

- We test many different factors: Wind Speed, Humidity, Traffic Fumes (NO2), and Ozone.
- We test them over different time delays (lags) to see if a spike in traffic in the morning actually *causes* a spike in dust by lunch time.

Why is this interesting?

Math is complicated, and normal users don't want to read complex formulas. So, we built a translator. Our engine takes all these massive math calculations and turns them into simple, plain-English advice.

Instead of just showing "AQI 300", our app can proudly say: *"The air is bad today, and our engine proves it is being caused by heavy traffic fumes and trapped by low wind speeds."* This makes our project incredibly unique.

3. Another Cool Feature: Smart Forecasting

Standard projects use basic AI to guess a single number for tomorrow's pollution.

Our project uses a much smarter AI called a **Temporal Fusion Transformer**. The interesting part is that instead of giving just one guess, our AI gives a "Confidence Band." This means it shows a range of possibilities on the chart, which honestly tells the user how sure or unsure the AI is about the future weather.

4. Future Enhancement: The AI Chatbot

While our math and forecasting engines are fully built and working, we want to make the app even easier for normal people to use.

Our biggest future enhancement that we are currently working on is a Smart AI Chatbot.

In the future, this chatbot will:

- **Understand Locations:** You can ask it "Is it safe to go to the park?" and it will automatically find the nearest sensor.
- **Give Personal Advice:** If you tell the bot you have asthma, it will read our math formulas and give you clinical advice on whether you should wear a mask.
- **Speak in Local Languages:** We are working on making the bot actually speak out loud in Hindi, Tamil, and Kannada so anyone in India can use it without needing to read complex charts.

By combining advanced causality math with an easy-to-use platform, Aerolytics is doing much more than just showing a number on a screen.